

OSSP (Operating Systems And Systems Programming)

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SEC-8

SKILL-1:

Aim: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile To Analyze process abstraction, execute fork(), understand exec() family, analyze parentchild relationships, inspect process tree, practice system call tracing

```
yagnesh_katukala@yagnesh:~$ mkdir ossp-shell
cd ossp-shell
yagnesh_katukala@yagnesh:~/ossp-shell$ mkdir src include bin
yagnesh_katukala@yagnesh:~/ossp-shell$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
yagnesh_katukala@yagnesh:~/ossp-shell$ sudo apt install tree -y
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 122
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 2s (35.0 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42298 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
yagnesh_katukala@yagnesh:~/ossp-shell$
```



```

|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x62d2e3a4d000) = 0x62d2e3a4d000
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_ypi
xel=0}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_ypi
xel=0}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
openat(AT_FDCWD, ".", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY
) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
newfstatat(AT_FDCWD, ".", {st_mode=S_IFDIR|0755, st_size=4096,
...}, 0) = 0
getdents64(3, 0x62d2e3a4e150 /* 13 entries */, 32768) = 384
getdents64(3, 0x62d2e3a4e150 /* 0 entries */, 32768) = 0
close(3) = 0
write(1, "Makefile build exec_demo inc"... , 100Makefile b
uild exec_demo include shell tests
README.md docs fork_demo parent_child src
) = 100
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NU
LL) = 0
munmap(0x779fa2420000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++

```


Code:

```
#include <stdio.h>

int main()
{
    printf("Linux Shell Project Started!\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if(pid == 0)
        printf("I am Child Process\n");
    else
        printf("I am Parent Process\n");

    return 0;
}
```

```
#include <unistd.h>

int main()
{
    execl("/bin/ls", "ls", NULL);
    return 0;
}
```

